



ITC 221 - APPLICATION DEVELOPMENT AND EMERGING TECHNOLOGIES
PROJECT PROPOSAL

TEAM MEMBERS:

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PROJECT TITLE:	Medical Clinic Notify+: Appointment Management
PROJECT DESCRIPTION: Medical Clinic Notify+ is a web-based application designed to streamline appointment scheduling between patients and clinic staff while ensuring secure identity verification and efficient communication through AI and cloud-based technologies. Patients can register, log in with OTP authentication, update profiles, and book appointments by selecting their preferred date and healthcare provider. The system integrates an AI-Assisted ID Verification System, allowing users to upload or capture their ID for validation. AI-powered OCR extracts and verifies key details such as name, ID number, and expiration date. If unclear or mismatched data is detected, the ID is flagged for manual admin review to enhance security and prevent fraud. Patients can track their appointment status (Pending, Approved, Denied, or Completed) and receive automated SMS for confirmations, reminders, and updates, reducing missed appointments. Administrators have tools to manage and approve appointments, verify patient identities, monitor assigned schedules, and generate reports to optimize clinic operations. As a cloud-based platform, Medical Clinic Notify+ runs on browsers like Google Chrome and Mozilla Firefox, allowing access from any device without installation. The system ensures secure, real-time updates and efficient clinic operations, addressing challenges like staff unavailability or scheduling conflicts. By integrating AI-powered ID verification, OTP security, automated notifications, and cloud storage, Medical Clinic Notify+ enhances patient engagement, minimizes fraud risks, and improves overall healthcare service delivery.	
PROJECT FEATURES:	
1. User Authentication The system ensures security for both users and administrators by requiring registered accounts to access the platform. This feature prevents unauthorized access, ensuring that only verified users can book or manage appointments within the system.	6. Search Functionality The system has a search feature that helps users find information easily. Students can look up their past and upcoming appointments by entering details like the appointment date, doctor’s name, or service type. Administrators can use it to find student records, appointment requests, and medical history. This makes it quicker and more convenient to access important data.
2. Password Hashing and Session Management	7. Downloadable Appointment Summary



To enhance security, the system implements password hashing, which encrypts user passwords to protect against unauthorized access. Additionally, session management ensures that users remain logged in securely while preventing session unauthorized usage of accounts.	To help students keep track of their medical visits, the system allows them to download a PDF summary of their appointment details. This document serves as a soft copy for their reference and helps them monitor their health records within the school clinic.
3. CRUD Operation The system follows CRUD (Create, Read, Update, Delete) operations to Manage user accounts and appointments. Users can create an account and book an appointment, while administrators can read and review submitted appointment requests. Users also have the ability to update their personal information if needed. The delete function allows users to cancel an appointment, but only if the administrator has not yet approved it.	8. Status Tracking The system provides real-time updates on appointment statuses, ensuring that students are always informed about their appointment progress. The statuses include Pending, Approved, Denied, or Completed, allowing users to stay updated on any changes to their scheduled appointments.
4. Booking Appointment Students can book an appointment by selecting an available doctor, date, and time that suits their schedule. Additionally, they can choose the specific service they require from the clinic, such as medical certification, physical exams, or other health-related services. This feature ensures a structured and hassle-free booking process for students.	9. Basic Profile Management User profiles are generated based on the information provided during registration. Students can update their details if necessary, ensuring that their records remain accurate. This feature helps the clinic maintain up-to-date information for more efficient appointment management.
5.Fully responsiveness for any device. The system is designed to be fully responsive, ensuring compatibility across all devices, including desktops, tablets, and mobile phones.	10. Dashboard Statistics Administrators have access to a dashboard that provides a statistical overview of clinic activities. This includes the total number of appointments, pending requests, and completed consultations.



This feature allows students and administrators to access the platform conveniently from any device, providing a seamless user experience regardless of screen size.		The statistical data helps clinic staff monitor trends and improve overall clinic operations.	
TECHNOLOGIES TO BE USED:			
1. Automated Communication Systems and Cloud-Based Messaging APIs. Firestore Database – Stores user details, ID number, verification status, and logs. Firestore Storage – Stores the uploaded ID image.		2. Web Server: XAMPP (Software)	
2. Database Software:		4. Text Editor: Visual Studio code (Software)	
5. Browser: Chrome (Software)		6. <i>Programming languages</i> :, HTML, CSS, JavaScript, PHP or Python	
7. <i>Design: Figma or Canva</i> (Software)		8. CSS Framework: Bootstrap (Software)	
9. Internet Connection (Hardware)		10. Operating system: Windows 10 (Software)	
11. EMERGING TECHNOLOGY The AI-Assisted ID Verification System enhances security by allowing users to upload or capture an ID image for verification. The system utilizes Firestore as its cloud database to store user data securely. For ID verification, the system integrates Google Cloud Vision API (AI-powered OCR) to extract and validate ID details automatically. If inconsistencies, missing information, or potential fraud are detected, the ID is flagged for manual review to ensure accuracy. For added security, Firebase Authentication with OTP verification is required upon login, preventing unauthorized access. By combining AI-powered OCR, Firestore for cloud storage, and OTP-based authentication, the system ensures an efficient, secure, and fraud-free identity verification process.			